

Siddhesh Gunjal

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EDUCATION

University of Mumbai

Bachelor's Degree, Mechanical

Agnel Polytechnic, Vashi

Diploma, Mechanical

Navi Mumbai, India

June 2015 – May 2018

Navi Mumbai, India

June 2012 – April 2015

EXPERIENCE

NielsenIQ

Senior Research Scientist

Pune, India

July 2024 – Present

Leading research and innovation to enhance AI systems powering the world's most trusted consumer intelligence platform, measuring and analyzing over 97% of global GDP. Built AI systems that save \$750K+ annually in operational costs while leading revenue-generating projects contributing \$100K+ in new annual revenue, transforming data into actionable insights for top-tier clients worldwide.

Tech-stack: Python, PyTorch, JavaScript, Eliza (in-house framework), DSPy, Crawl4AI, vLLM, Flask, Gunicorn, AzureML, Docker, GitHub workflow, CUDA/Cudnn

Apaar Information systems

ML Engineer

Navi Mumbai, India

Oct 2022 – May 2024

Built gstGPT (GenAI powered GST solution for businesses). Built Emo-Detect [1], a user emotion tracking application through CCTV & personal computer to analyse on-floor behaviour. Deployed Multiple WebAR Apps with custom ML based image tracking on client side to increase engagement by 80% and lead conversion by 40% [2] [3]. Deployed NLP based invoice processor to save around 70% time & minimize human error in invoice processing.

Tech-stack: Python, Dart, JavaScript, SQL, PyTorch, TensorFlow, Flask, FastAPI, ElasticSearch, MongoDB, AWS, Docker, GitHub workflow, CUDA/Cudnn, TensorRT

upGrad

Adjunct Faculty | Data Science Coach

Mumbai, India

Mar 2020 – Mar 2023

As a Data Science Coach my work is to deliver sessions to small groups of people explaining them core concepts of the ML and AI, solving doubts, guiding them to solve actual world problems with Data [4].

Topics covered: Machine Learning, Computer Vision, Natural Language Processing, Reinforcement Learning

LivePixel Technologies

ML & XR Developer

Navi Mumbai, India

Jul 2020 – Oct 2022

Deployed multiple MR Experiences with nReal MR glasses [5] for Science Museum (Government project) to increase student engagement by 90%. Developed motion control & NLP based voice control software for 3D holograms for interaction with digital character [6]. AR Car showroom app to experience and shop automotive from anywhere [7]. RatLab VR app to allow medical students learn operation procedures and increase their confidence by 40% during actual procedure/practical [8].

Tech-stack: Python, JavaScript, C/C++, C#, PHP, SQL, AWS, CUDA/Cudnn, TensorRT, Unity Engine, Unreal Engine 4 & 5, SolidWorks, AutoCad, Blender

Bombay Stock Exchange Limited

Visiting Professor

Mumbai, India

Aug 2019 – Mar 2020

Taught Advanced Data Analytics for stock market and create live dashboards with R-shinny. Tableau integration with R-server for advanced analytics.

Topics covered: Machine Learning with python, Data visualisation with R (R-Shinny), Tableau, R-server integration with Tableau

riidl (Research Innovation Incubation Design Laboratories)

Teaching Assistant | Professor: Vikramank Singh

Mumbai, India

Feb 2018 – Jul 2018

Taught implementation of various projects using various deep learning models like AI in Self Driving Car (Vanilla model by Nvidia), Q-learning in simple Atari game (Checkers & Flappy Bird), Data visualization and produced a document to debug the code of AI in Self-driving car.

OPEN-SOURCE CONTRIBUTIONS

Slackker

Author & Maintainer | Website: [Link](#) | GitHub: [Slackker](#)

Slackker sends real-time notifications for any Python pipeline, ML training or automation workflow with metrics and optional attachments for better analysis.

FluxTest

Author & Maintainer | Website: [Link](#) | GitHub: [FluxTest](#)

FluxTest is a self-hosted server network diagnosis tool for YOUR infrastructure (home lab, VPS, bare-metal, edge nodes, private cloud).

Aya 101 & Aya Expanse

Contributor | HuggingFace: [Dataset](#), [Aay 101](#), [Aya Expanse](#)

The Aya model is a multilingual LLM that follows instructions in 101 languages. Aya Expanse is a successor of Aya 101 with. More capable & context length upto 128K!

TableQA

Contributor | GitHub: [tableQA](#)

AI Tool for querying natural language on tabular data. Built using QA models from transformers.

Spatial AR on Web

Author & Maintainer | Live WebApp: [Link](#) | GitHub: [SpatialAR](#)

This WebApp utilizes WebXR API to communicate with Android's ARCore for spatial computing of the surroundings to detect planes and objects

SKILLS

Programming Languages: Python, C/C++, C#, JavaScript, HTML/CSS, Dart, R, PHP

Machine Learning: PyTorch, TensorFlow, Keras, LoRA Fine-tuning, Reinforcement Learning, Computer Vision, Natural Language Processing

Technical Skills: CUDA/Cudnn, TensorRT, ONNX Runtime, Tableau, Flutter, React, BabelJS, Docker, IBM Watson, Amazon Web Services (AWS), Google Cloud Platform (GCP), CircleCI, Unity Engine, Unreal Engine 4 & 5, SolidWorks, AutoCad, Catia, Blender

TALKS & MEETUPS

Keynote Speaker on "AI/ML for Mechanical Contractors" hosted by Institution of Engineers India (IEI) [9]

Judge at "DataHack 2.0" organized by DJS S4DS hosted by Dwarkadas J. Sanghvi College of Engineering [10]

Industry Expert Talk on "Deep Learning – A revolution in Mechanical and Automobile Industry" hosted by Pillai College of Engineering [11]

Speaker on "Artificial Intelligence workshop" hosted by Indian Institute of Technology, Kanpur (IIT Kanpur) – Techkriti 2020 (Non-Core weekend) [12]

Hands-on workshop on "Deep learning in Computer Vision" hosted by 91springboard [13]

Keynote Speaker on "Q-Learning in ML and Game Development Demo" at Big Data Mumbai v 14.0 – Come to code [14]

Keynote Speaker on "Intelligent bots and Self-driving Cars" at Big Data Mumbai v12.0 [15]